

Yash Rawal

+91-9399159685 | yashrawal987@gmail.com | linkedin.com/in/rawal-yash | github.com/YashRL | Ratlam (457001), Madhya Pradesh, India

Professional Summary

Applied AI & Machine Learning Engineer serving a **1M+ user scale** by architecting enterprise agentic platforms, hierarchical RAG architectures, and automated intelligence pipelines across 120+ organizations. Specialized in Python, LLM orchestration (LangGraph, CrewAI, ReAct loops), vector search, and model evaluation control planes. Active contributor to Spring AI, bridging generative AI and model fine-tuning with resilient production systems.

Skills

- Languages & Backend:** Python, SQL, Java, FastAPI, Spring Boot, PostgreSQL, REST APIs
- AI & ML Engineering:** Agentic AI, RAG & Vector Search, Multi-Agent Systems, Context Graphs, Semantic Search
- AI & ML Frameworks:** LangGraph, CrewAI, LangChain, Spring AI, PyTorch, vLLM, LlamaIndex, GROBID
- Platforms & Models:** Azure OpenAI, Claude, Amazon Bedrock, Google Vertex AI, Meta Llama, Qwen
- Infrastructure & Security:** Docker, Kubernetes, Linux, CI/CD, LLM-as-Judge, Red Teaming, PII Sanitization

Work Experience

Bigscal Technologies Pvt. Ltd.
AI/ML Engineer
Remote

Oct 2024 - Present

- Deployed a production-scale agentic platform and multi-agent systems serving **1M+ active ERP users** across 120+ enterprise tenants.
- Designed a capability contract layer to execute nested profiles inside a single ReAct loop, reducing execution overhead by **3x** and token consumption by 70%.
- Architected multi-agent routing and copilot workflows with LangGraph and CrewAI, deploying **STREAMABLE HTTP MCP servers** for real-time tool discovery and sub-second execution.
- Developed the Prompt Lab evaluation control plane to collect reasoning trajectories and human preference annotations, establishing a scalable **Human Feedback (RLHF) Pipeline**.
- Implemented a hierarchical RAG pipeline indexing 10,000+ technical documents using GROBID, pdfplumber, and PostgreSQL/PGVector, achieving a **90%+ Recall@10** retrieval rate.
- Secured agentic workflows against OWASP LLM vulnerabilities using phantom-token context propagation, runtime tool monitoring, and trained token-classification models for **automated PII redaction**.

Bigscal Technologies Pvt. Ltd.
AI/ML Engineer Intern
Surat, India

Apr 2024 - Sep 2024

- Engineered clinical summarization pipelines for 116K+ medical records with 4-bit quantized Llama 3, and designed real-time WebSocket audio streaming for Indic ASR (Whisper, Vosk, NeMo).
- Researched audio digital signal processing (STFT, Mel-spectrograms, MFCC feature extraction via librosa/torchaudio) and CNN audio architectures (VGGish, ResNet), compiling custom Indic LJSpeech-format datasets.

Projects

Modular Agent Harness & Memory System ("Brain") | Open Source
Principal Creator & Architect of an open-source plug-and-play agentic framework designed as modular building blocks for flexible enterprise workflows, implementing a formal ReAct loop with step budgets, vector tool routing, and dynamic MCP settings caching.
Java, Spring Boot, PostgreSQL, WebSockets

Spring AI (spring-projects/spring-ai) | Open Source
Merged PR #5711 to propagate reasoningContent from AWS Bedrock and OpenAI-compatible models, and resolved Converse API prompt-caching TTL latency consistency.
Java, Spring Boot, Python

Mini Language Model (~85M Parameter Architecture) | Personal Research
Pretrained an ~85M-parameter transformer to ~1.43 validation loss on TinyStories; engineered FragmentStream chunked attention to eliminate $O(T^2)$ VRAM overhead on commodity GPUs and custom Triton kernels with KV caching for low-latency inference.
Python, PyTorch, CUDA, Triton, Tesla P100/T4

Education

Master of Computer Applications (Artificial Intelligence & Machine Learning) - Parul University

2022 - 2024

Certifications

Microsoft Certified: Azure AI Fundamentals (AI-900)	Microsoft
Microsoft Certified: Azure Fundamentals (AZ-900)	Microsoft